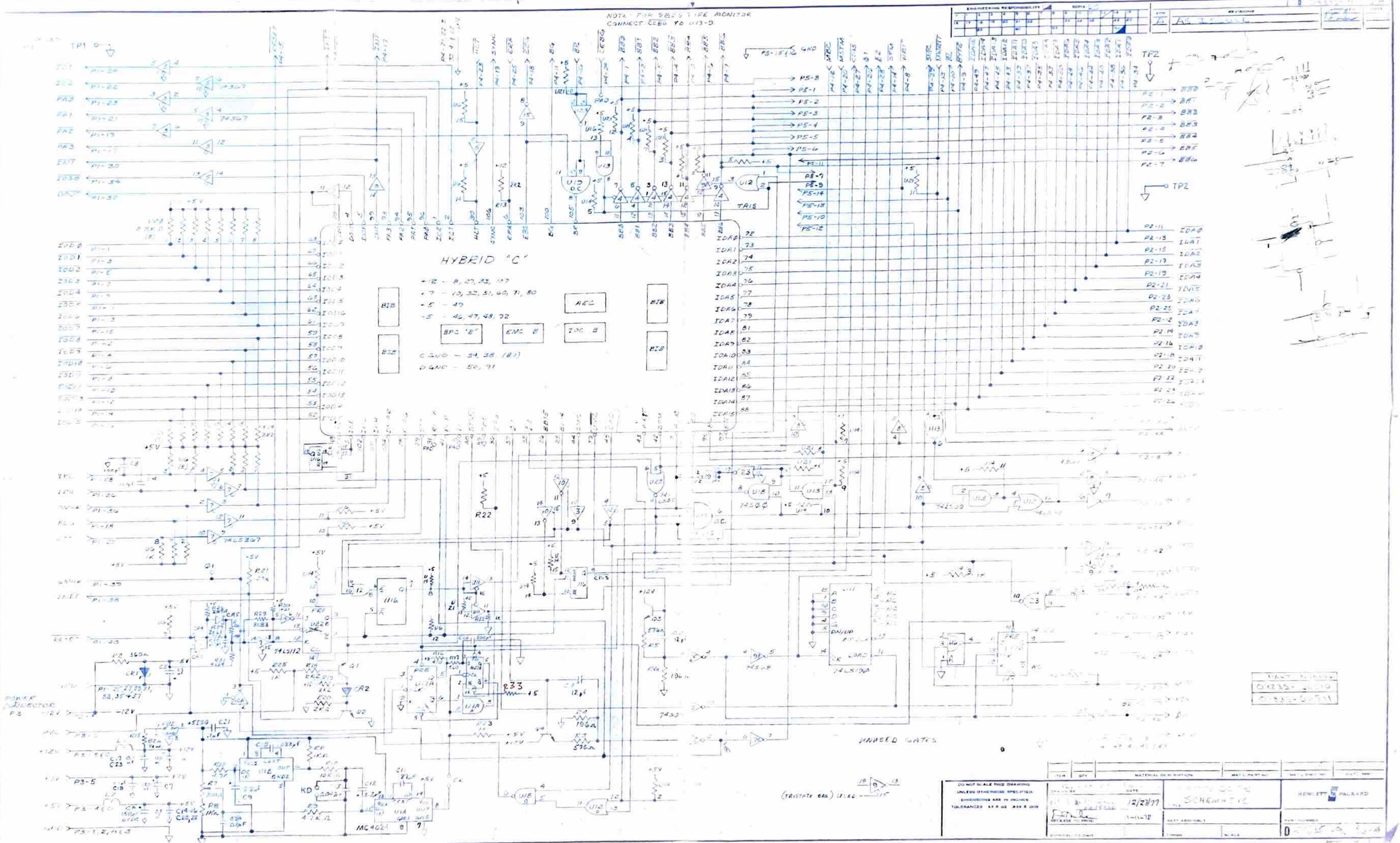
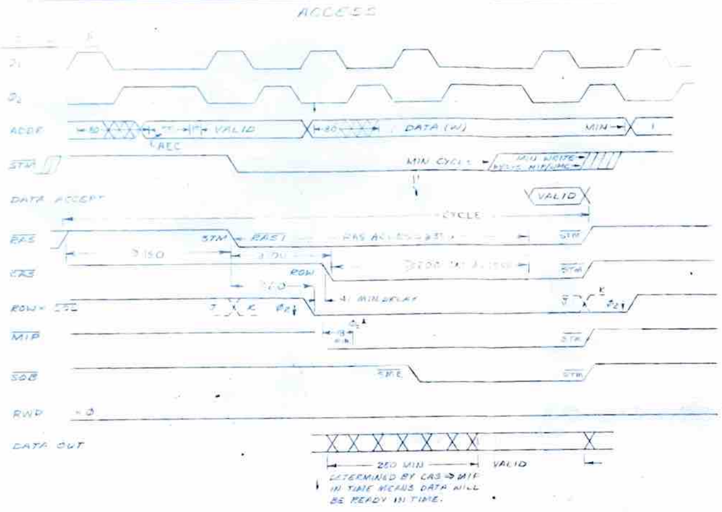

NOTE: FOR 9825 TIRE MONITOR
CONNECT CEBG TO U13-9
HYBRID "C"
+12 – 8, 27, 33, 107
+7 – 10, 32, 51, 60, 71, 80
+5 – 49
-5 – 46, 47, 48, 92
C GND – 34, 38, (87)
D GND – 50, 91
BIB
BIB
AEC
BIB
BIB
BPC "B"
EMC B
IOC B
TP1
TP2
TP2
POWER CONNECTOR P3
74367
74367
74LS367
74LS112
74LS00
74LS190
74S00
7400
MC4024
R22
R33
UNUSED GATES
(TRISTATE GATES) U120
PART NUMBER
01235-

DO NOT SCALE THIS DRAWING
UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES .XX ± .02 .XXX ± .005

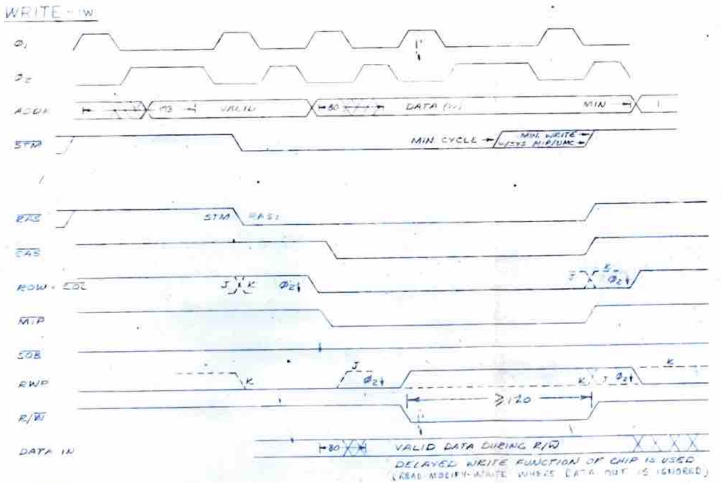
ITEM | QTY | MATERIAL DESCRIPTION | MAT'L PART NO.

DRAWN BY | DATE 12/27/77
RELEASE TO PROD. | 1-13-78
TITLE SCHEMATIC
NEXT ASSEMBLY
SCALE
HEWLETT PACKARD
PART NUMBER
D

ACCESS


φ1
φ2
ADDR
VALID
DATA (W)
MIN
tAEC
STM
MIN CYCLE
MIN WRITE
W/SYS MIP/UMC
DATA ACCEPT
VALID
CYCLE
RAS
STM
RAS1
≥150
CAS
ROW
ROW = COL
J K φ2↓
41 MIN DELAY
MIP
SOB
RWP = φ
DATA OUT
250 MIN
VALID
DETERMINED BY CAS ⇒ MIP
IN TIME MEANS DATA WILL
BE READY IN TIME.

WRITE - IW

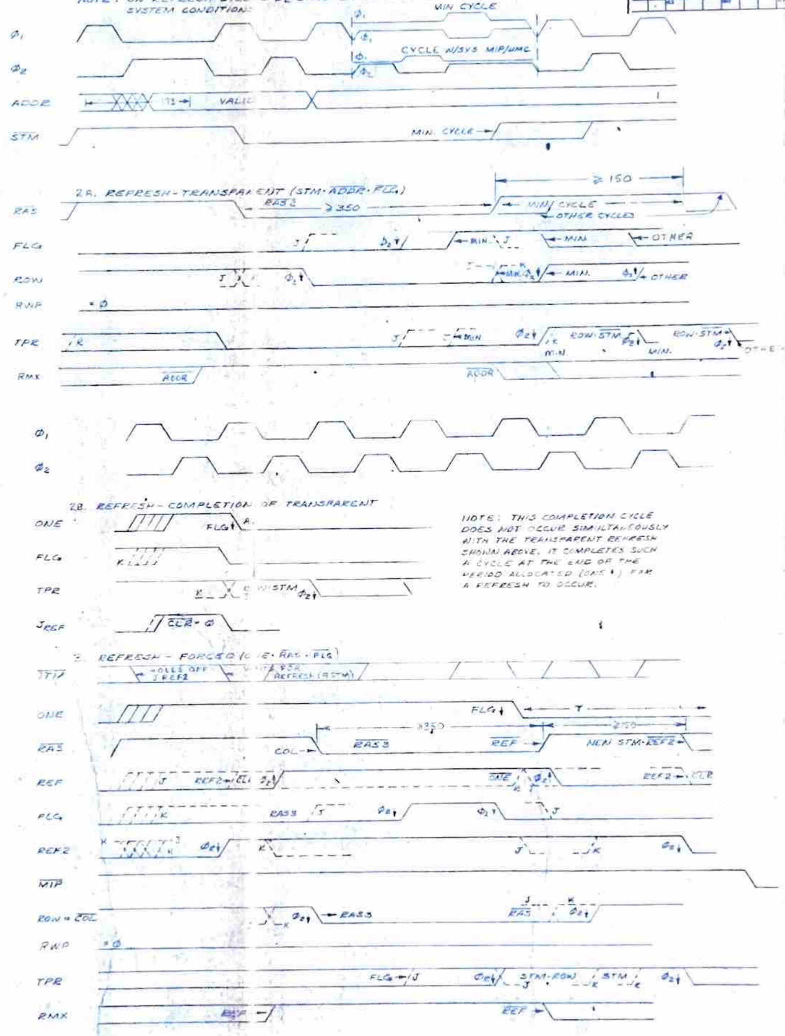

φ1
φ2
ADDR
VALID
DATA (W)
MIN
STM
MIN CYCLE
MIN WRITE
W/SYS MIP/UMC
RAS
STM
RAS1
CAS
ROW = COL
J K φ2↓
MIP
SOB
RWP
K
J
φ2↓
R/W
≥120
DATA IN
80
VALID DATA DURING R/W

DELAYED WRITE FUNCTION OF CHIP IS USED
(READ-MODIFY-WRITE WHERE DATA OUT IS IGNORED)

NOTES:
1. ADDR. IS LATCHED FROM STM↓ TO STM↑
2. TPR ELIMINATES A 2ND TSP ON THE SAME STM.
3. FLG INDICATES TSP HAS OCCURRED AND FORCED REFRESH IS UNNECESSARY.
4. TPR PROP TO ENABLE RAS SHOULD BE S GUARANTEED STM OR RAS WILL SHORTEN AFTER FORCED REFRESH.
5. REFRESH MUST GO AHEAD FOR STM PRESENT TO ANOTHER SYSTEM ADDRESS.
6. MEMORY MUST NOT BE EXPECTED TO INDEFINITELY HOLD DATA ACCESSED. (REFRESH CANNOT OCCUR FOR SUCH A CONDITION)

REFRESH

NOTE: ON REFRESH CYCLES φ2 STRETCH IS CONTROLLED BY OTHER SYSTEM CONDITIONS.


φ1
φ2
MIN CYCLE
CYCLE W/SYS MIP/UMC
ADDR
VALID
STM
MIN. CYCLE
2A. REFRESH - TRANSPARENT (STM·ADDR·FLG)
≥150
RAS
RAS2
≥350
MIN CYCLE
OTHER CYCLES
FLG
MIN
OTHER
ROW
RWP = φ
TPR
ROW·STM
RMX
ADDR
2B. REFRESH - COMPLETION OF TRANSPARENT
ONE
FLG
TPR
JREF
CLR·φ
3. REFRESH - FORCED (ONE·RAS·FLG)
HOLDS OFF JREF2
REFRESH (RSTM)
ONE
FLG↓
T
≥350
≥150
RAS
COL
RAS3
REF
NEW STM·REF2
REF
REF2
CLR
FLG
RAS3
REF2
MIP
ROW = COL
RAS3
RWP = φ
TPR
FLG → J
STM·ROW
RMX
REF

NOTE: THIS COMPLETION CYCLE DOES NOT OCCUR SIMULTANEOUSLY WITH THE TRANSPARENT REFRESH SHOWN ABOVE. IT COMPLETES SUCH A CYCLE AT THE END OF THE PERIOD ALLOCATED (ONE↓) FOR A REFRESH TO OCCUR.

FLIPFLOP EQUATIONS

	CLOCK
$J_{COL} = K_{ROW} = RAS$ $K_{COL} = J_{ROW} = \overline{RAS}$	$\phi_2\downarrow$
$J_{RWP} = COL \cdot WRITE \cdot RAS1$ $K_{RWP} = \overline{RAS}$	$\phi_2\downarrow$
$J_{FLG} = RAS3 + RAS2 \cdot COL$ $K_{FLG} = ONE$	$\phi_2\downarrow$
$J_{REF} = ONE$ $\overline{CLR}_{REF} = REF2$ $K_{REF} = \overline{ONE}$	$\phi_2\downarrow$
$J_{REF2} = ONE \cdot \overline{RAS} \cdot FLG$ $K_{REF2} = \overline{REF} \cdot ROW$	$\phi_2\downarrow$

MONOSTABLE

$B_{trig} = 1$

$A_{trig} = FLG\downarrow$ OUTPUT

$RC = 14.7\,\mu sec$

$\frac{\times 128}{1.88\ msec}$

$J_{TPR} = FLG$ $K_{TPR} = \overline{FLG} \cdot STM \cdot ROW$	$\phi_2\downarrow$

COUNT REF = { FLG↓ FOR ½ LOAD (TOP HALF ONLY)
CT↓ FOR FULL LOAD

SIGNAL EQUATIONS

$RAS = RAS1 + RAS2 + RAS3$

$R/W = \overline{READ} \cdot RWP \cdot RAS1$

$MEB = STM \cdot ADDR \cdot \overline{COL \cdot REF2}$ (ONLY APPLIES TO 9825 RETROFIT - NOT USED IN RAVEN)

$MIP = \overline{CAS}$

ROW ENABLE = ROW

REF ENABLE = $\overline{ADDR} + REF$

$RAS1 = ADDR \cdot \overline{\overset{RSTM}{STM \cdot \overline{REF2}}}$

$RAS2 = \overline{ADDR} \cdot TPR \cdot STM \cdot \overline{REF2} \cdot (\overline{FLG} \cdot ROW)$

$RAS3 = REF \cdot COL$

OUTPUT ENABLE = $\overline{SOB} \cdot RAS1 = \overline{SOB} \cdot CAS$

$RMX = \overline{ADDR} + REF = \overline{ADDR \cdot \overline{REF}}$ (REFRESH ADDRESS SELECTED)

$RMX \cdot \overline{ADDR}$ ENABLES RAS2 (TRANSPARENT REFRESH)
$RMX \cdot REF$ ENABLES RAS3 (FORCED REFRESH)
$\overline{REF} \cdot ADDR$ ENABLES RAS1 (ACCESS)

$RSTM = STM \cdot \overline{REF2}$

$CS_{ENABLE} = \overline{REF} \cdot ADDR$

ITEM	QTY	MATERIAL DESCRIPTION	MAT'L PART NO	MAT'L DWG NO	MAT'L SPEC

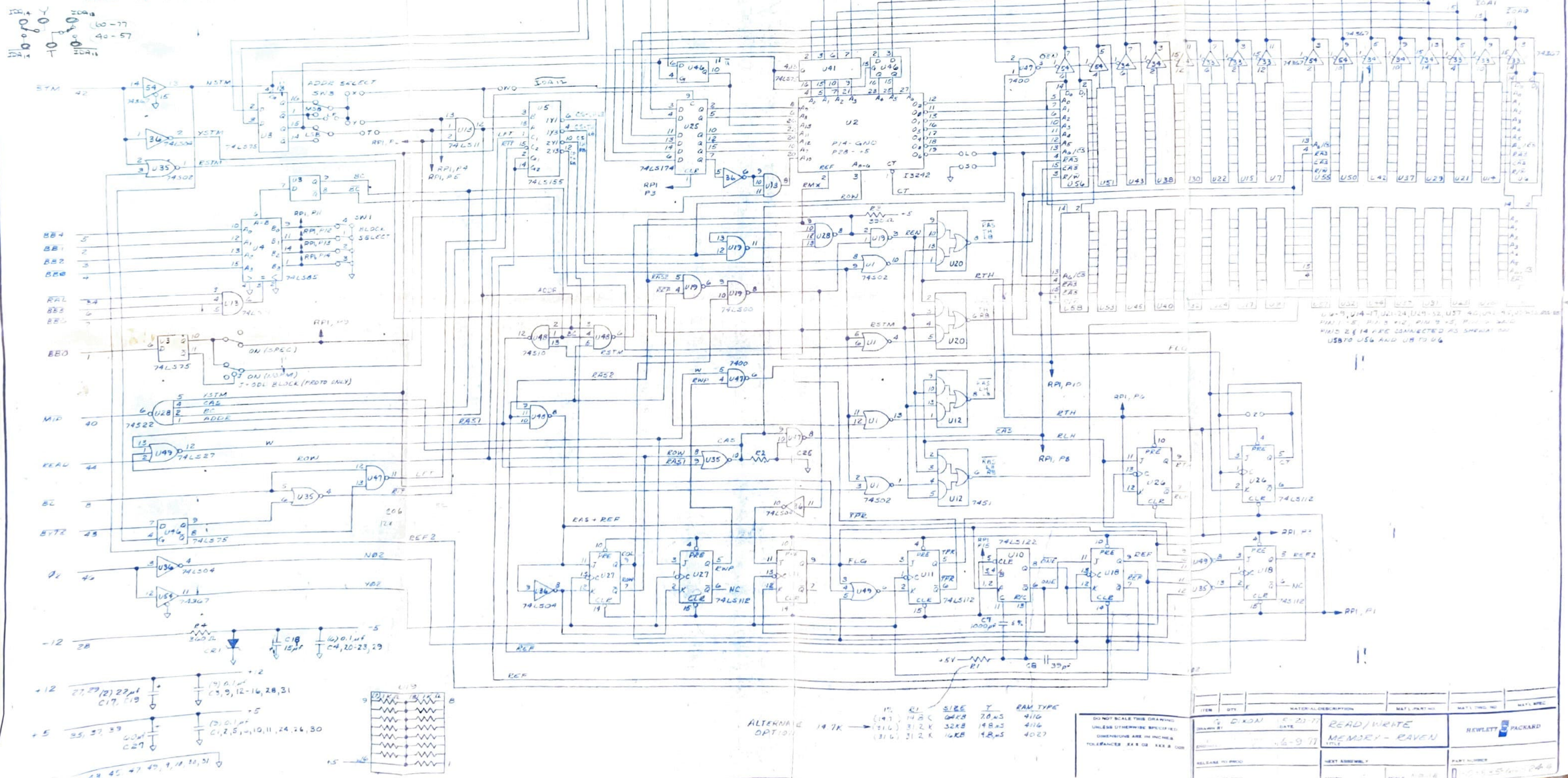
DO NOT SCALE THIS DRAWING
UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES .XX ± .02 .XXX ± .005

DRAWN BY: G. DIXON	DATE: 6-1-77	TITLE: TIMING - RAVEN READ/WRITE MEMORY	HEWLETT PACKARD
ENGINEER	6-9-77		
RELEASE TO PROD.		NEXT ASSEMBLY	PART NUMBER
SUPERSEDES DWG.		FINISH / SCALE	D-09835-66524-7

	LOAD SW3	LOAD U12,26	LF	W	JUMPER X	Y	Z	T	[illegible]	R1
[illegible]	YES	YES	NO	YES	NO	YES	NO	YES	YES	14.7KΩ
[illegible] 16-16K (POSITION TH0-15)	YES	NO	NO	NO	YES	YES	YES	NO	NO	31.6KΩ
[illegible]KB 32-16K	NO	YES	YES	NO	YES	NO	NO	NO	NO	31.6 KΩ

ALTERNATE OPTION:

32KB	16-16K (POSITION TH0-TH15)	YES	YES X12-[illegible]	NO	NO	YES	YES	NO	NO	NO	14.7KΩ


IDA14
Y
IDA13
60-77
40-57
T
STM
YSTM
ESTM
ADDR SELECT SW3
BLOCK SELECT SW1
BB4
BB1
BB2
BB0
RAL
BB3
BB5
BBO
ON (SPEC)
ON (ROM)
J-ODD BLOCK (PROTO ONLY)
MIP
READ
BC
BYTE
Φ2
-12
+12
+5
U2
P14 - GND
P28 - +5
I3242
U41
U25
U46
U5
U13
U3
U4
U28
U49
U35
U36
U54
U47
U19
U1
U20
U12
U48
U27
U11
U10
U18
U26
74LS04
74LS75
74LS85
74LS174
74LS155
74LS11
74LS00
74S02
74S10
74S22
74LS27
74LS112
74LS122
74367
7400
IOA15
IOA14
IOA13
IOA12
IOA11
IOA10
IOA9
IOA8
IOA7
IOA6
IOA5
IOA4
IOA3
IOA2
IOA1
IOA0
30B
RAS
CAS
RTH
RLH
REF
REF2
FLG
RMX
ROW
CT
RPI, P10
RPI, P8
RPI, P1
RPI, P5
R4 360Ω
CR1
C18 15μf
(6) 0.1μf C4, 20-23, 29
(2) 22μf C17, C19
(9) 0.1μf C3, 9, 12-16, 28, 31
(9) 0.1μf C1, 2, 5, 6, 10, 11, 24, 26, 30
60μf C27
C7 1000pf
+5V R1 C8 39pf

U6-9, U14-17, U21-24, U29-32, U37-40, U42-45, [illegible]
PIN 1 -5, PIN 8 +12, PIN 9 +5, PIN [illegible] GND
PINS 2 & 14 ARE CONNECTED AS SHOWN ON
U58 TO U56 AND U8 TO U6

ALTERNATE OPTION 14.7K →

1%	R1	SIZE	T	RAM TYPE
(14.7)	14.8 K	64KB	7.0 nS	4116
(31.6)	31.2 K	32KB	14.8 nS	4116
(31.6)	31.2 K	16KB	14.8 nS	4027

DO NOT SCALE THIS DRAWING
UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES .XX ± .02 .XXX ± .008

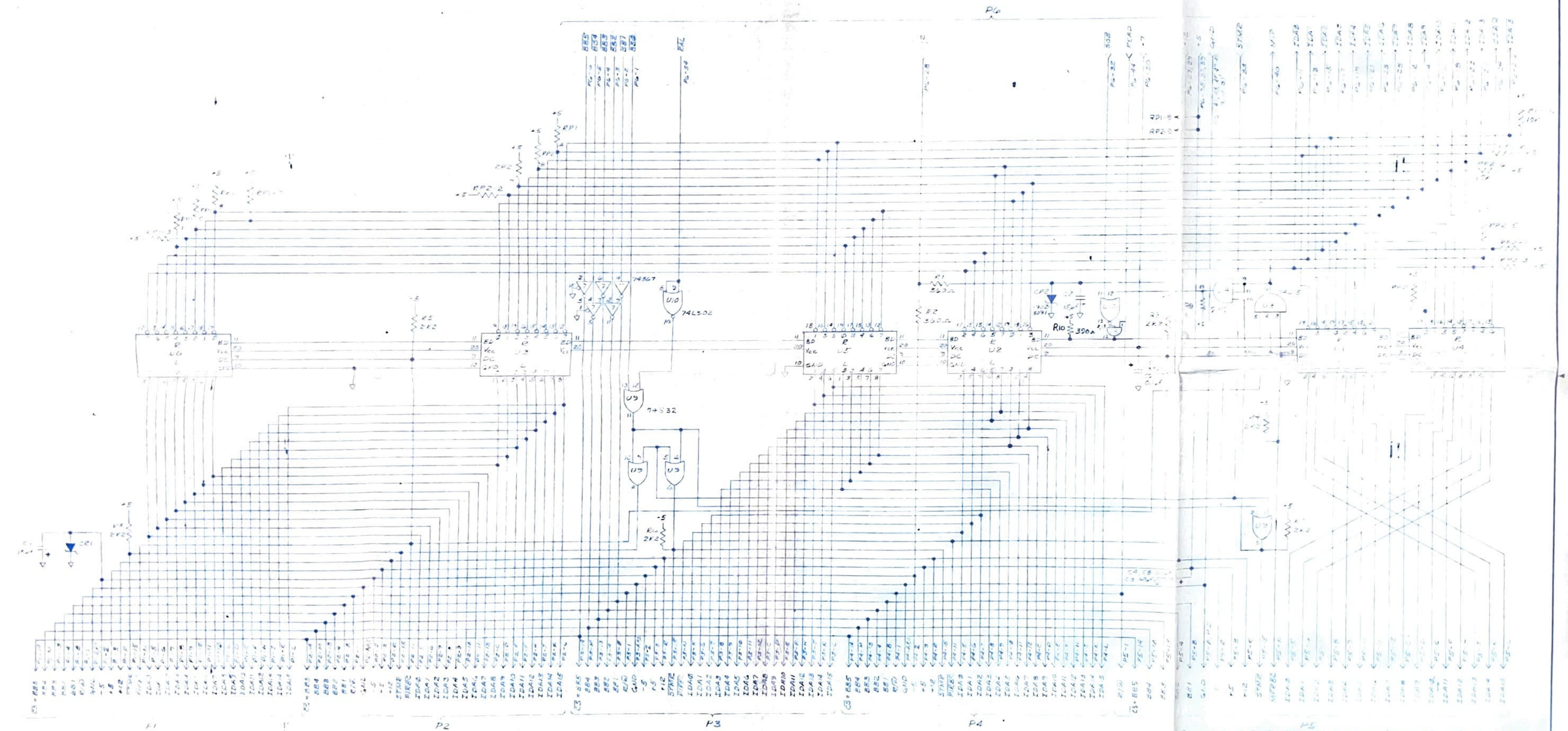
DRAWN BY G. DIXON
DATE 5-20-77
ENGINEER 6-9-77
TITLE READ/WRITE MEMORY - RAVEN
HEWLETT PACKARD
RELEASE TO PROD
NEXT ASSEMBLY
PART NUMBER [illegible]

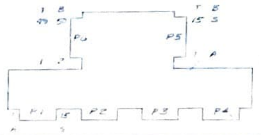
ENGINEERING RESPONSIBILITY

0	1	2	3	4	5	6	7	9	10	11	12	14	15
16	17	18	19	20	21	22		33	34	35		42	43
		63			90								

SEPIA

SYM	REVISIONS	APPROVED	DATE


P6
BB5
BB4
BB3
BB2
BB1
BB0
BC
P6-34
P6-28
P6-32
P6-44
P6-50
SOB
PLAD
STMR
M/P
+5
GND
RP1
RP2
RP2-2
RP1-5
RP2-5
74367
U10
74LS02
U9
74S32
R1
360Ω
R2
360Ω
CR2
1902
3391
C2
15μF
R10
390Ω
R3
2K2
R5
2K2
R4
2K2
R11
2K2
R7
2K2
U1
U2
U3
U4
U5
U6
U8
BD
VCC
DC
GND
CR1
C1
IDA0
IDA1
IDA2
IDA3
IDA4
IDA5
IDA6
IDA7
IDA8
IDA9
IDA10
IDA11
IDA12
IDA13
IDA14
IDA15
CS-BB5
CS-BB5
R/W
STRB
+12
-5
P1
P2
P3
P4


1 B
49 50
P6
T B
15 S
P5
1 2
1 A
P1
15
P2
P3
P4
T
B
A
S

REV
B.

P5

PURPOSE OF REV. B – ENABLE BIB TO XFER DATA TO PLUG IN R/W DURING WRITE. KH 2/2/77

U8-C UTILIZE UNUSED SECTION

ITEM	QTY	MATERIAL DESCRIPTION	MAT'L PART NO.	MAT'L DWG NO.	MAT'L SPEC
R10	1	RESISTOR	0683-3915		

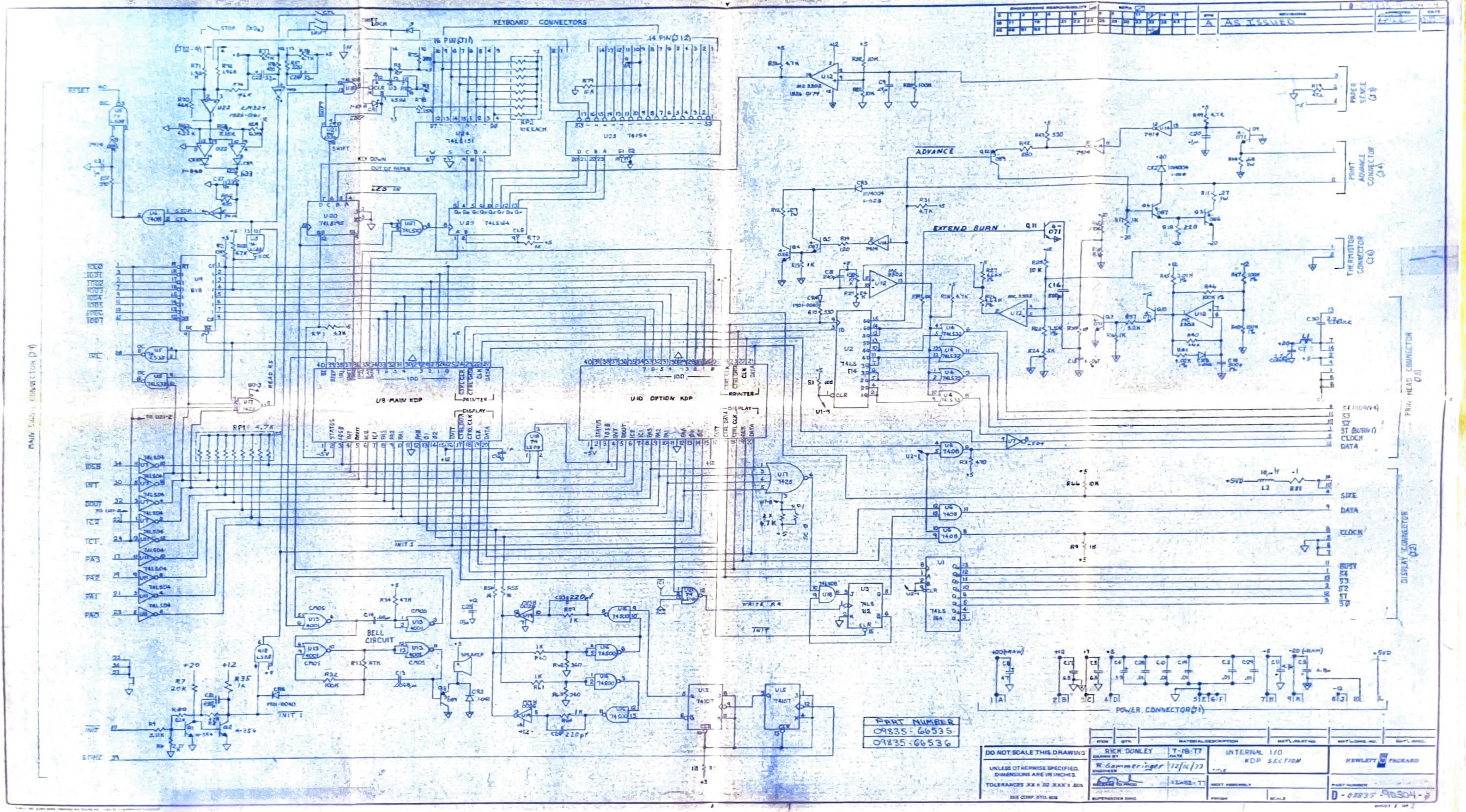
DO NOT SCALE THIS DRAWING
UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES XX ± .02 XXX ± .005

DRAWN BY: D. [illegible] DATE: 1-14-77
ENGINEER: 1/6/77
RELEASE TO PROD
SUPERSEDES DWG

TITLE: ROM INTERFACE PANEL

NEXT ASSEMBLY

HEWLETT PACKARD


KEYBOARD CONNECTORS
16 PIN (J11)
14 PIN (J12)
STOP (30a)
SHIFT LOCK
SHIFT
CTL
RESET
U24 74LS151
U25 74154
RP2 10K EACH
KEY DOWN
OUT OF PAPER
LED IN
U20 74LS195
U23 74LS164
U21 74LS10
U22 LM324
U19 8T8
MAIN SIGNAL CONNECTOR (J1)
IRL
READ R5
RP1 4.7K
IOSB
INT
DOUT
IC2
IC1
PA3
PA2
PA1
PA0
INIT
6 MHZ
74LS04
U8 MAIN KDP
U10 OPTION KDP
PRINTER
DISPLAY
STATUS
CTRL CLK
CTRL DATA
CLK
DATA
U17 74LS25
U18 74LS08
INIT 3
INIT 1
WRITE R4
BELL CIRCUIT
U13 4001
U15 4001
CMOS
SPEAKER
U16 74S00
U13 74107
U15 74107
ADVANCE
EXTEND BURN
MC 3302
U2 74LS 114
U4 74LS32
U6 7408
U7 LS04
U1 74LS 164
U3 74LS 112
PAPER SENSE (J3)
PRINT ADVANCE CONNECTOR (J4)
THERMISTOR CONNECTOR (J6)
PRINT HEAD CONNECTOR (J5)
S4 (BURN 4)
S3
S2
S1 (BURN 1)
CLOCK
DATA
DISPLAY CONNECTOR (J2)
SIZE
DATA
CLOCK
BUSY
S4
S3
S2
S1
S0
+5VD
POWER CONNECTOR (J1)

PART NUMBER
09835-66535
09835-66536

ENGINEERING RESPONSIBILITY

LTR	REVISIONS	APPROVED	DATE
A	AS ISSUED		

DO NOT SCALE THIS DRAWING

UNLESS OTHERWISE SPECIFIED. DIMENSIONS ARE IN INCHES.

TOLERANCES .XX ± .02 .XXX ± .005

SEE CORP. STD. 606

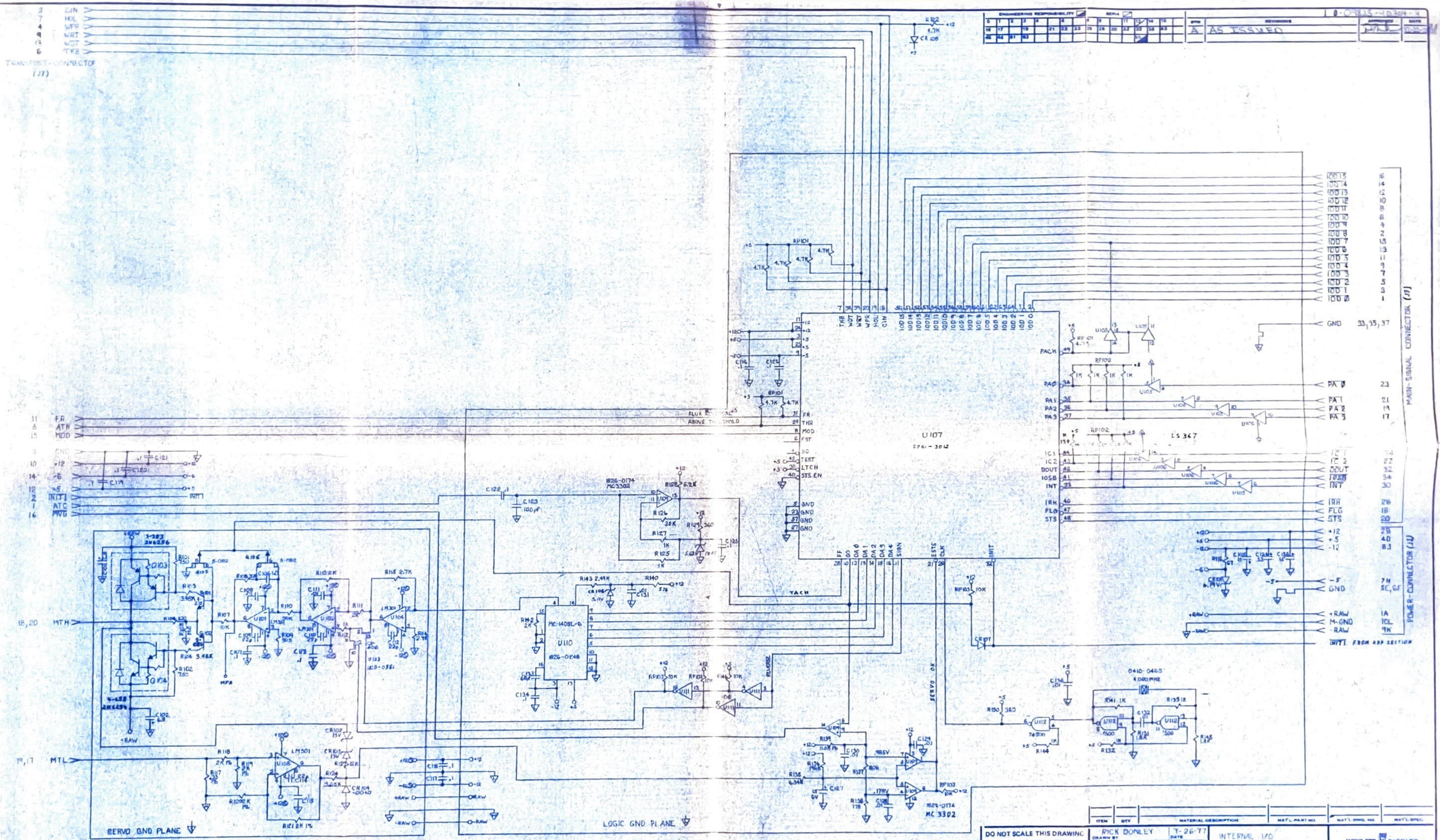
RICK DONLEY DRAWN BY	7-18-77 DATE	INTERNAL I/O KDP SECTION TITLE	HEWLETT PACKARD
R. Gammeringer ENGINEER	12/16/77		
RELEASE TO PROD.	12-22-77	NEXT ASSEMBLY	PART NUMBER
SUPERSEDED DWG.		FINISH / SCALE	D-09835-90304-A

SHEET 1 OF 2

ENGINEERING RESPONSIBILITY								SEPIA							
0	1	2	3	4		6		8	9		11	12	14	15	
16	17		19		21	22	23	25	28	30	32	33	38	43	
45	46	61	63									54			

D-09835-90304-4

LTR	REVISIONS	APPROVED	DATE
A	AS ISSUED		


3 CIN
7 HOL
4 WPR
9 WRT
13 WDT
6 TKB
TRANSPORT CONNECTOR (J7)
11 FR
6 ATR
15 MOD
GND
10 +12
14 -6
12 +5
2 INIT
1 ATC
16 MVG
18,20 MTH
19,17 MTL
SERVO GND PLANE
LOGIC GND PLANE
U107
5061-3012
U110
MC-1408L-6
1826-0248
FLUX
ABOVE THRESHOLD
TACH
SERVO OK
LS 367
0410-0465
4.000 MHZ
MC 3302
1826-0174
LM301
RP101
RP102
RP103
IOD 15
IOD 14
IOD 13
IOD 12
IOD 11
IOD 10
IOD 9
IOD 8
IOD 7
IOD 6
IOD 5
IOD 4
IOD 3
IOD 2
IOD 1
IOD 0
GND 33, 35, 37
PA 0 23
PA 1 21
PA 2 19
PA 3 17
IC 1 24
IC 2 22
DOUT 32
IOSB 34
INT 30
IRH 26
FLG 18
STS 20
MAIN-SIGNAL CONNECTOR (J1)
+12 2B
+5 4D
-12 8J
-5 7H
GND 5C, 6F
+RAW 1A
M-GND 10L
-RAW 9K
POWER-CONNECTOR (J1)
INIT1 FROM ADP SECTION

ITEM	QTY	MATERIAL DESCRIPTION	MAT'L PART NO.	MAT'L DWG. NO.	MAT'L SPEC.

DO NOT SCALE THIS DRAWING

UNLESS OTHERWISE SPECIFIED.
DIMENSIONS ARE IN INCHES.

TOLERANCES .XX ± .02 .XXX ± .005

SEE CORP. STD. 600

RICK DONLEY DRAWN BY	7-26-77 DATE	INTERNAL I/O TACO SECTION TITLE	HEWLETT PACKARD
R. Gemmeringer ENGINEER	12/16/77		
RELEASE TO PROD.	12-22-77	NEXT ASSEMBLY	PART NUMBER D-09835-90304-4
SUPERSEDES DWG.		FINISH	SCALE

SHEET 2 OF 2